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Algorithms and Probability **Peer Sheet P1** **FS 2026**

The solutions for this sheet must be submitted on Moodle by 12 March, 15:59. Afterwards you are assigned solutions of another student, which you have to grade (and upload to Moodle) until 11:59 of the following TUESDAY.

Exercise P1.1 – *Matchings and magic tricks*

(a) We say that a bipartite graph $G = (A \cup B, E)$ is (k, ℓ)-regular if every vertex in A has degree exactly k, and every vertex in B has degree exactly ℓ. Show that when $k \geq \ell$, any (k, ℓ)-regular graph G has a matching of size exactly $|A|$.

(b) A magician together with their assistant performs the following trick. An audience member picks 5 arbitrary cards from a standard 52-card playing deck, and gives those cards to the magician's assistant. The assistant then picks four out of the five cards, and shows those cards to the magician, in order of the assistant's choice. The magician now, with certainty, announces what is the remaining fifth card in the assistant's hand, not seen by the magician at any point.

How could the magician and the assistant (who both have unlimited brain-memory) have chosen a strategy ahead of time so that they would succeed in this trick, regardless of which five cards are selected by the audience member? Once the show has started, the magician and the assistant cannot communicate in any way outside of the scope of the trick (i.e. the only communication from the assistant to the magician is the sequence of the 4 cards the assistant presents to them in the order of their choice). *Hint:* Use subproblem (a).